Interaction Between Formal Education and Structural Brain Atrophy in Alzheimer's Disease

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Background

- Educational attainment is believed to contribute to cognitive reserve (CR), a factor that helps maintain cognitive functioning despite age-related decline (Liu et al., 2024).
- Dementia can cause structural changes in the brain, such as progressive brain atrophy (Lv et al., 2025).
- Research showed mixed results, with some studies showing that education is linked to slower cognitive decline but also may make the individual more sensitive to atrophy (Mungas et al., 2018).
- This study used secondary quantitative analysis of the OASIS dataset (N = 350) to examine whether years of educational attainment directly predict whole brain volume atrophy (nWBV) in dementia patients
- We hypothesized that higher educational attainment is associated with higher baseline brain volumes

Methods

Software: All data processing, statistical modeling, and visualization (Figures 1-4) were conducted using **R Studio**.

Secondary analysis utilized the **OASIS-1 cross-sectional MRI dataset (N=350)**. **Figure 1** established a baseline using the full lifespan cohort (18-96 years), while **Figures 2-4** utilized the clinical dementia sub-group.

Normalized Whole Brain Volume (nWBV) was selected as the primary outcome. The normalization process mathematically controls for individual variations in intracranial volume.

Age was utilized as a primary control in all education-based models to ensure observed trends were associated with education

Gender and **Socioeconomic Status (SES)** were addressed as potential confounding variables and analyzed to ensure that the educational association was not skewed by sex-based differences or financial background.

Inclusion/Exclusion Criteria: Final analysis focused on Clinical Dementia Rating (CDR) stages 0 -1. Participants categorized as CDR 2.0 (Moderate) were excluded (n=3)

A **One-Way ANOVA** was initially conducted to compare mean **nWBV** across clinical diagnosis groups (Demented vs. Nondemented) to establish the primary structural impact of the disease

Statistical Modeling: A **Two-Way ANOVA** evaluated the interaction between educational attainment and clinical diagnosis on brain volume (nWBV).

Results

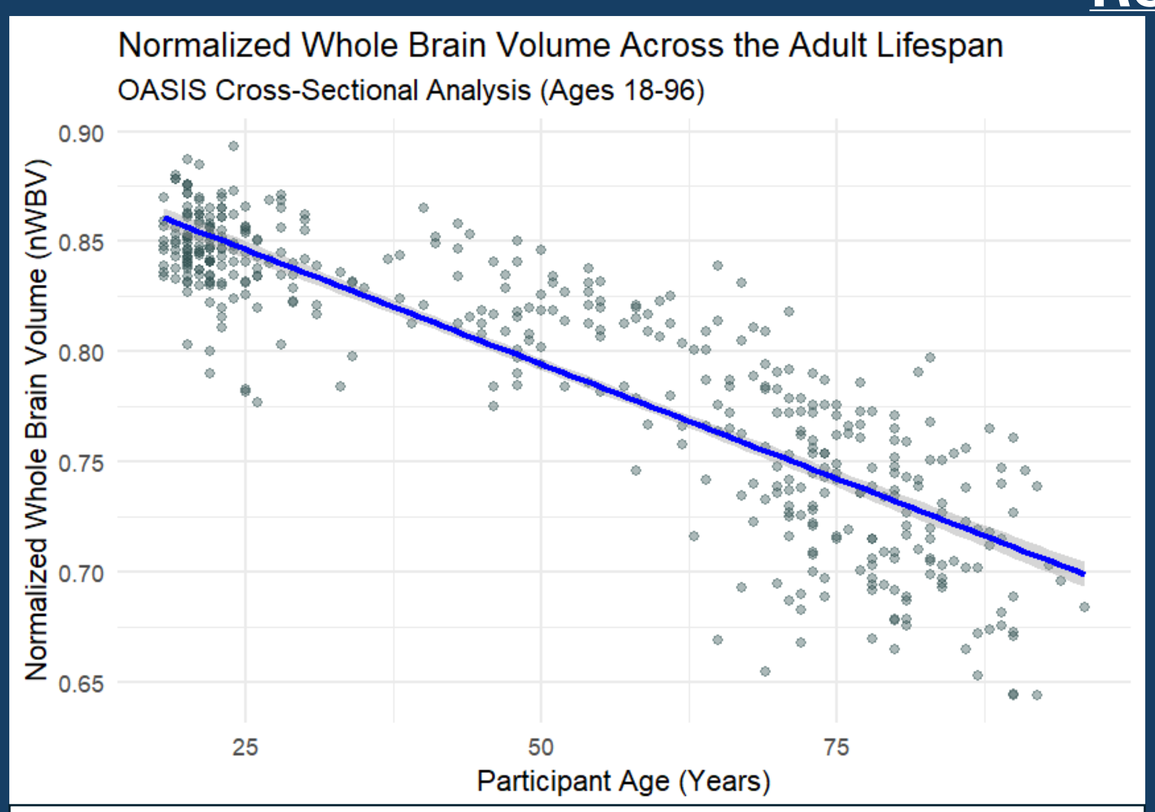


Figure 1. Baseline normalized whole brain volume (nWBV) that predictably declines as a function of age across the adult lifespan ($p < 0.001$, $R^2 = 0.764$).

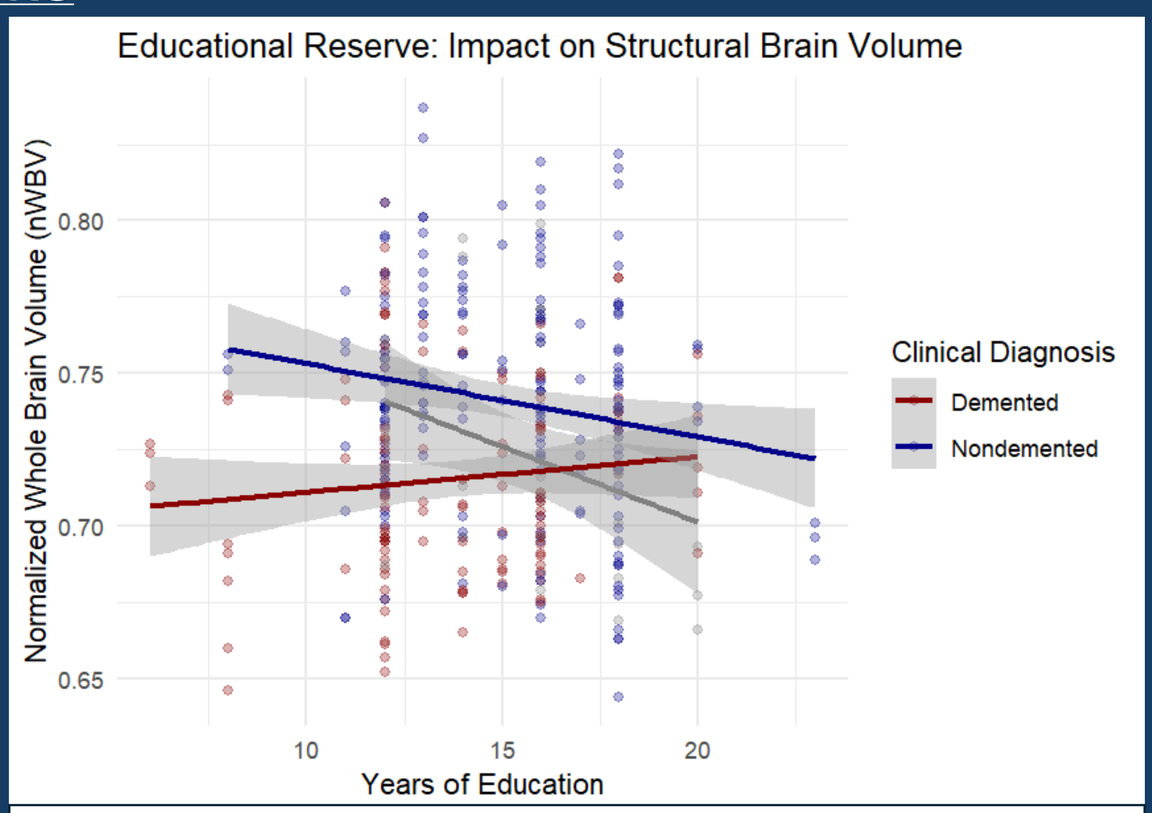


Figure 4. This plot demonstrates how years of formal education correlate with brain volume. The distinct regression slopes for both groups, illustrating that higher educational attainment may provide a buffer to brain atrophy as a clinical symptom ($p = 0.0115$, $n^2 = 0.067$).

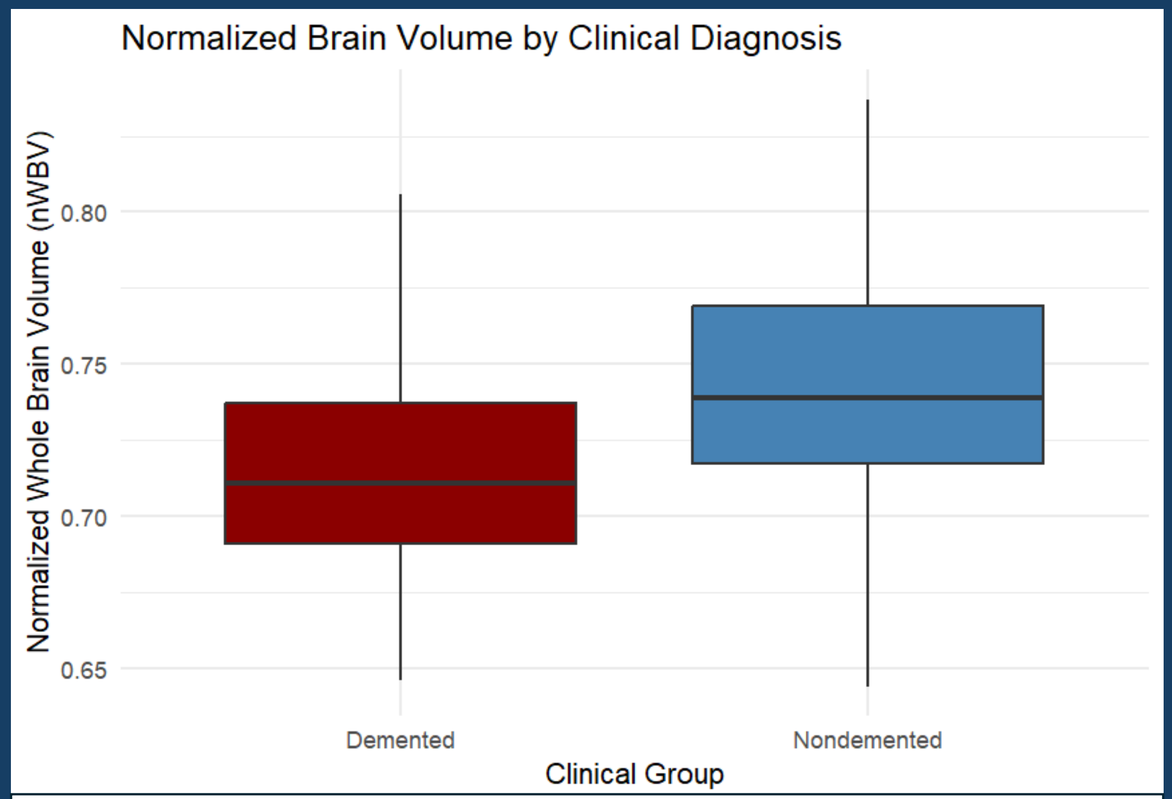


Figure 2. Boxplot comparing physical brain volume of participants categorized by clinical diagnosis. One way ANOVA confirmed a highly significant reduction in nWBV with demented compared to nondemented participants ($p < 0.001$, $n^2 = 0.101$).

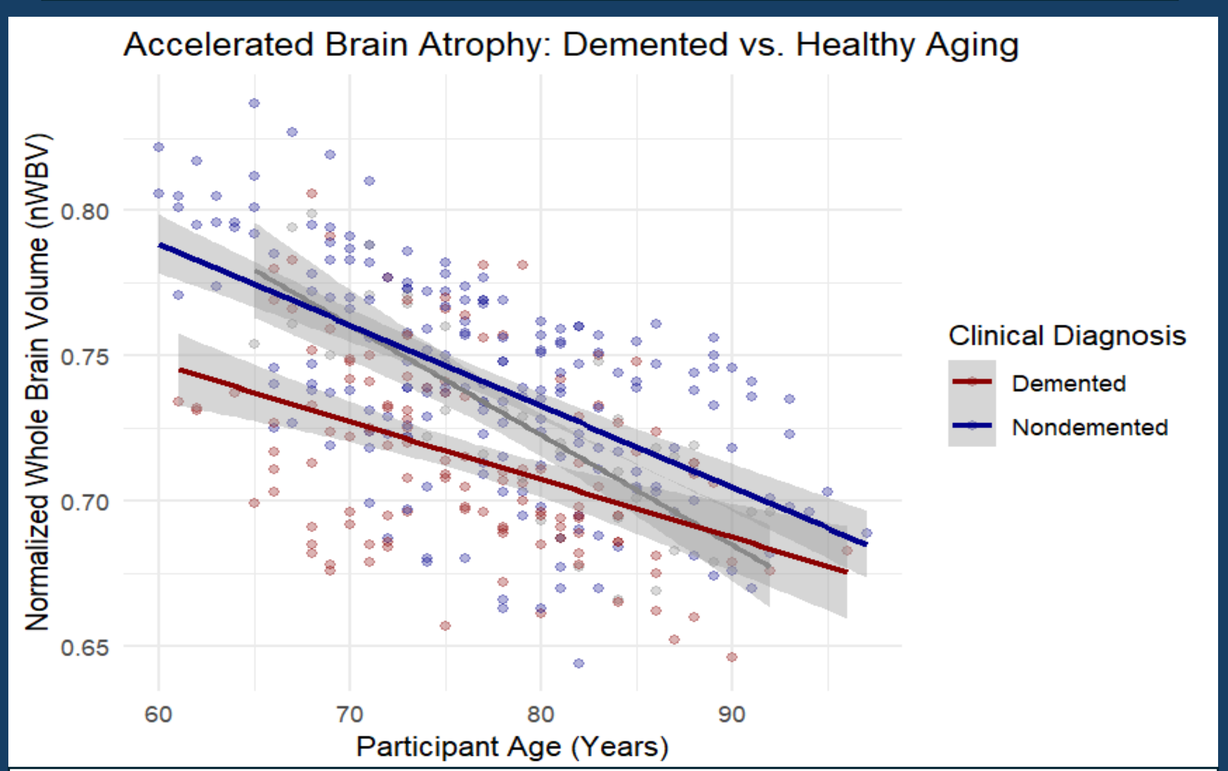


Figure 3. Brain volume loss (nWBV) between demented and nondemented seniors. While age related decline is universal, the demented cohort (red line) shows a significantly lower volume compared to healthy cohorts (blue line).

Conclusion

- Two-Way ANOVA confirmed a significant interaction between education and clinical diagnosis ($p = 0.0115$, $n^2 = 0.067$), providing evidence of the correlation between education and brain volume varying across clinical stages (Figure 4).
- Results are consistent with Cognitive Reserve Hypothesis, suggesting formal education as a structural buffer (Stern, 2009), maintaining higher nWBV
- While age related nWBV is universal (Figure 1), the clinical cohort shows a significant structural drop ($p < 0.001$) and accelerated decline across senior lifespan (Figure 3), consistent with progressive brain atrophy and structural changes by dementia (Lv et al,. 2025).
- Future studies should combine structural MRI data with emerging technologies like blood-based biomarkers and Transcranial Magnetic Stimulation (TMS) to identify early synaptic dysfunction (Koch & Spampinato, 2022).
- Ultimately, evidence suggests being cognitively active early in life is associated with greater dementia prevention (Livingston et al., 2024), highlighting the correlation between education and increased structural resilience.

References

Koch, G., Spampinato, D (2022). Alzheimer disease and neuroplasticity. *Handbook of Clinical Neurology, 184,* 473-479. https://doi.org/10.1016/B978-0-12-819410-2.00027-8

Liu, Y., Lu, G., Liu, L., He, Y., & Gong, W. (2024). Cognitive Reserve over the life course and risk of dementia: A systematic review and meta-analysis. Frontiers in Aging Neuroscience, 16. https://doi.org/10.3389/fnagi.2024.1358992

Livingston, G., Huntley, J., Liu, K. Y., Costafreda, S. G., Selbæk, G., Alladi, S., Ames, D., Banerjee, S., Burns, A., Brayne, C., Fox, N. C., Ferri, C. P., Gitlin, L. N., Howard, R., Kales, H. C., Kivimäki, M., Larson, E. B., Nakasujja, N., Rockwood, K., ... Mukadam, N. (2024). Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. The Lancet, *404*, 572–628. https://doi.org/10.1016/S0140-6736(24)01296-0

Lv, L., Guo, H., Zhao, Z., & Zhao, X. (2025). Structural and microstructural changes in white and gray matter across the Alzheimer's disease continuum. Frontiers in Aging Neuroscience, 17. https://doi.org/10.3389/fnagi.2025.1693840

Mungas, D., Gavett, B., Fletcher, E., Farias, S. T., DeCarli, C., & Reed, B. (2018). Education amplifies brain atrophy effect on cognitive decline: Implications for Cognitive Reserve. Neurobiology of Aging, 68, 142–150. https://doi.org/10.1016/j.neurobiolaging.2018.04.002

Stern, Y. (2009). Cognitive reserve. Neuropsychologia, *47*(10), 2015–2028. https://doi.org/10.1016/j.neuropsychologia.2009.03.004

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